AMFIINTEL TERMINAL - Project Documentation

Enterprise Mutual Fund Intelligence & Portfolio Analytics Platform

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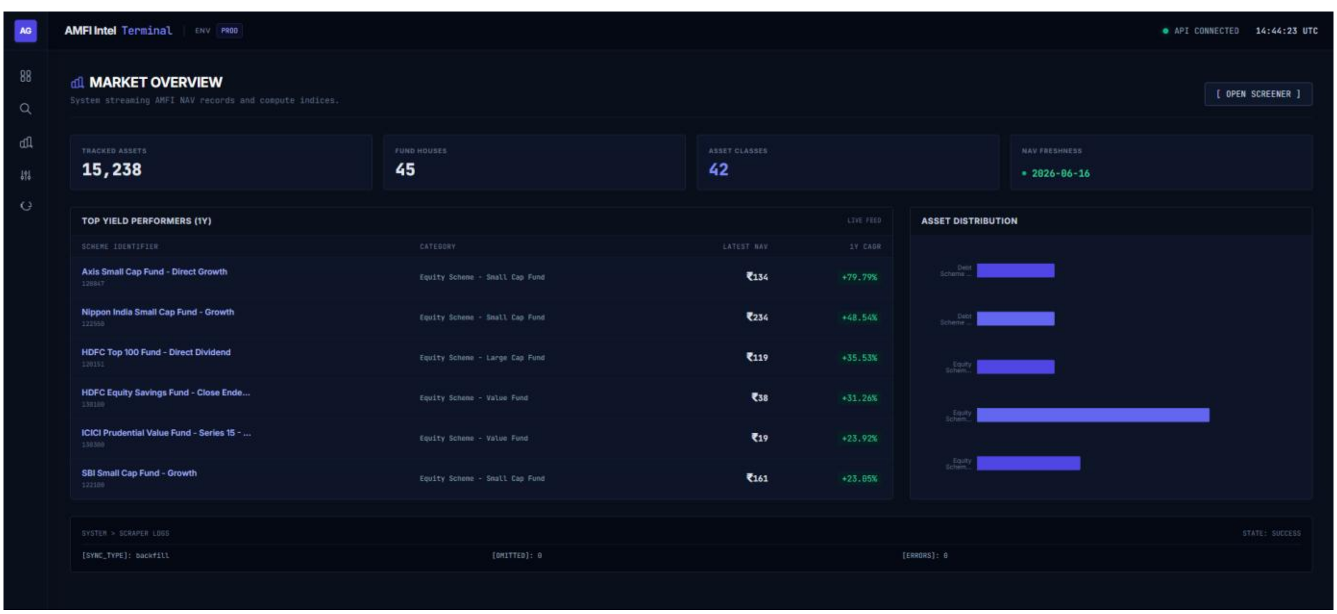
Abstract

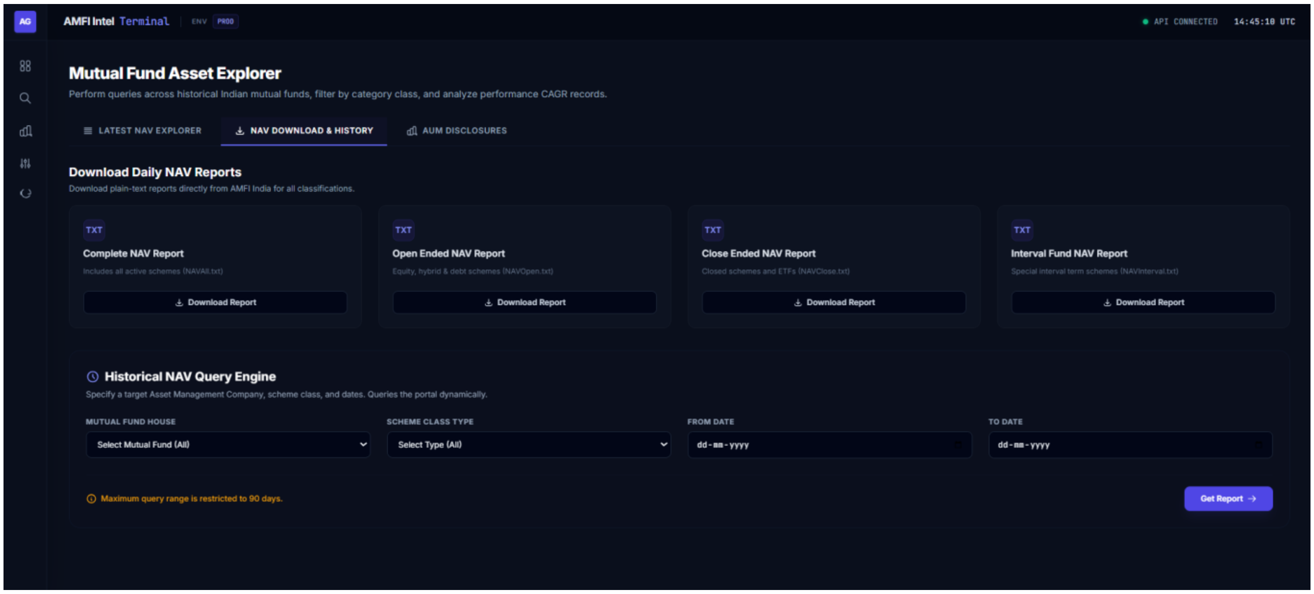
AMFIIntel Terminal is a production-grade Mutual Fund Intelligence and Portfolio Analytics Platform that transforms publicly available AMFI data into actionable investment insights. The platform automates data collection, historical NAV tracking, analytics generation, risk assessment, and portfolio analysis through a centralized fintech dashboard, enabling users to conduct comprehensive mutual fund research from a single interface.

Built using Next.js, FastAPI, PostgreSQL, and automated data engineering pipelines, the platform provides advanced capabilities such as fund comparison, portfolio overlap detection, diversification analysis, historical trend visualization, and professional report generation. By combining financial analytics with modern visualization and scalable backend architecture, AMFIIntel delivers a complete investment research ecosystem for investors, analysts, and financial professionals.

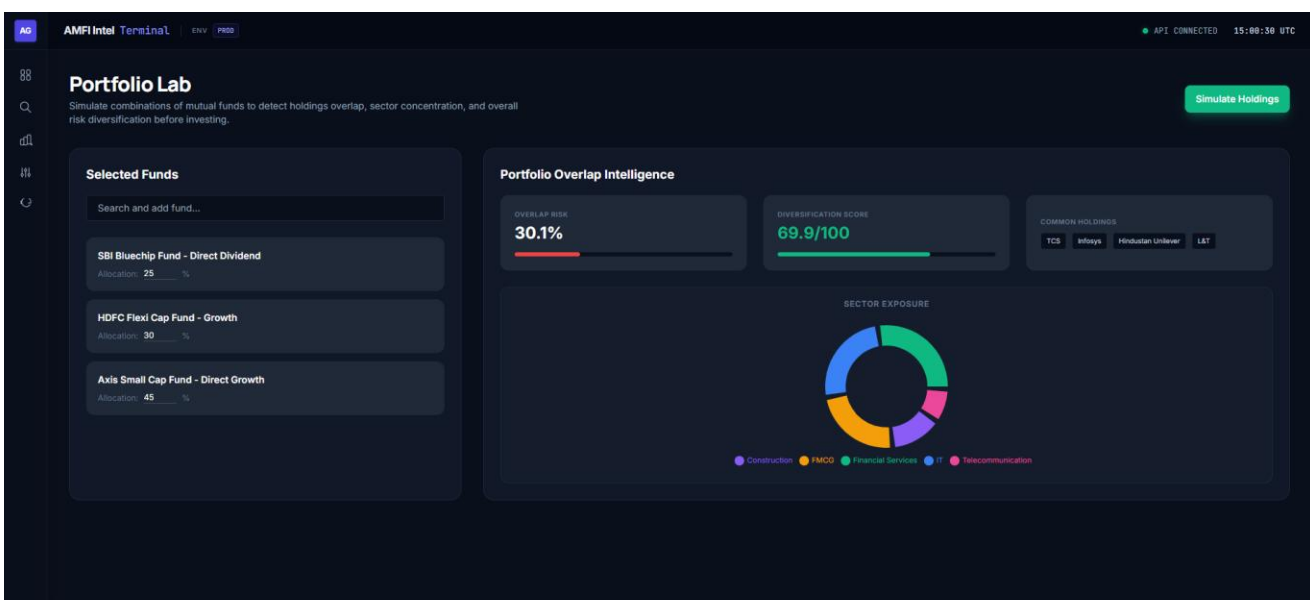
Introduction

AMFIIntel Terminal is a production-grade Mutual Fund Intelligence and Portfolio Analytics Platform designed to transform publicly available AMFI data into actionable investment insights. The platform automates data collection, historical NAV tracking, financial analytics, portfolio overlap detection, risk assessment, and report generation through a unified fintech dashboard. By combining data engineering, backend services, analytical processing, and modern visualization techniques, AMFIIntel enables investors and analysts to conduct comprehensive mutual fund research, performance evaluation, and portfolio analysis from a single platform.




AG
AMFI Intel Terminal
ENV PROD
API CONNECTED 14:45:10 UTC
Mutual Fund Asset Explorer
Perform queries across historical Indian mutual funds, filter by category class, and analyze performance CAGR records.
LATEST NAV EXPLORER
NAV DOWNLOAD & HISTORY
AUM DISCLOSURES
Download Daily NAV Reports
Download plain-text reports directly from AMFI India for all classifications.
TXT
Complete NAV Report
Includes all active schemes (NAVAll.txt)
Download Report
TXT
Open Ended NAV Report
Equity, hybrid & debt schemes (NAVOpen.txt)
Download Report
TXT
Close Ended NAV Report
Closed schemes and ETFs (NAVClose.txt)
Download Report
TXT
Interval Fund NAV Report
Special interval term schemes (NAVInterval.txt)
Download Report
Historical NAV Query Engine
Specify a target Asset Management Company, scheme class, and dates. Queries the portal dynamically.
MUTUAL FUND HOUSE
Select Mutual Fund (All)
SCHEME CLASS TYPE
Select Type (All)
FROM DATE
dd-mm-yyyy
TO DATE
dd-mm-yyyy
Maximum query range is restricted to 90 days.
Get Report


AG
AMFI Intel Terminal
ENV PROD
API CONNECTED 14:45:56 UTC
SBI Bluechip Fund - Direct Dividend
Equity Scheme - Large Cap Fund
SBI Bluechip Fund - Direct Growth
Equity Scheme - Large Cap Fund
SBI Healthcare Opportunities Fund - Growth
Equity Scheme - Sectoral/ Thematic
SBI Liquid Fund - Growth
Debt Scheme - Liquid Fund
SBI Magnum Midcap Fund - Growth
Equity Scheme - Mid Cap Fund
SBI Monthly Interval Fund - Plan A - Growth
Debt Scheme - Interval Fund
SBI Nifty 50 Index Fund - Growth
Other Schemes - Index Funds
SBI Nifty Index Fund - Growth
Equity Scheme - Large Cap Fund
SBI Small Cap Fund - Growth
Equity Scheme - Small Cap Fund
SBI Tax Advantage Fund - Series III - Growth
Equity Scheme - Value Fund
UTI Nifty 50 Index Fund - Growth
Other Schemes - Index Funds
₹10,000 COMPOUND INVESTMENT GROWTH SERIES
SBI Magnum Midcap Fund - Growth
Axis Small Cap Fund - Direct Dividend
SBI Small Cap Fund - Growth
SBI Bluechip Fund - Direct Dividend
₹16000
₹12000
₹8000
₹4000
₹0
RISK & RETURN COMPARISONS
Axis Small Cap Fund - Direct Dividend
1Y Returns:
+16.56%
Volatility:
+28.76%
Sharpe Ratio:
0.25
Max DD:
-40.51%
SBI Bluechip Fund - Direct Dividend
1Y Returns:
-8.16%
Volatility:
+13.51%
Sharpe Ratio:
0.07
Max DD:
-18.84%
SBI Magnum Midcap Fund - Growth


AG
AMFI Intel Terminal
ENV PROD
API CONNECTED 15:00:30 UTC
Portfolio Lab
Simulate combinations of mutual funds to detect holdings overlap, sector concentration, and overall risk diversification before investing.
Simulate Holdings
Selected Funds
Search and add fund...
SBI Bluechip Fund - Direct Dividend
Allocation: 25 %
HDFC Flexi Cap Fund - Growth
Allocation: 30 %
Axis Small Cap Fund - Direct Growth
Allocation: 45 %
Portfolio Overlap Intelligence
OVERLAP RISK
30.1%
DIVERSIFICATION SCORE
69.9/100
COMMON HOLDINGS
TCS
Infosys
Hindustan Unilever
L&T
SECTOR EXPOSURE
Construction
FMCG
Financial Services
IT
Telecommunication

Problem Statement

The Indian mutual fund ecosystem comprises thousands of schemes across multiple Asset Management Companies (AMCs), categories, and investment strategies. Although data is publicly available through AMFI, AMC websites, and financial research portals, it remains fragmented and difficult to analyze efficiently. Investors often spend significant time gathering information from multiple sources, resulting in inconsistent research processes and limited visibility into fund performance, risk, and market trends.

Additionally, most platforms focus on displaying current fund data while offering limited historical analysis, portfolio intelligence, and reporting capabilities. Investors struggle to evaluate long-term performance, compare funds across key metrics, identify portfolio overlaps, and assess diversification risks. The absence of integrated analytics and research tools makes informed investment decision-making both complex and time-consuming.

Project Vision

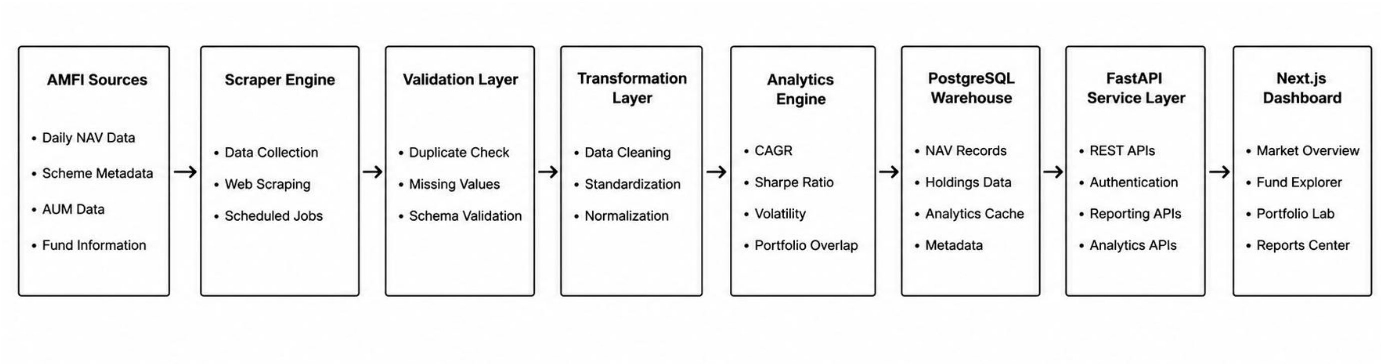
The vision of AMFIIntel Terminal is to create a comprehensive Mutual Fund Intelligence Platform that transforms fragmented financial data into meaningful, actionable investment insights. The platform aims to bridge the gap between raw mutual fund information and informed decision-making by providing investors, analysts, and researchers with a centralized ecosystem for fund discovery, performance evaluation, risk assessment, and portfolio analysis.

To achieve this vision, the platform focuses on:

- **Automated Data Intelligence through continuous synchronization and processing of AMFI mutual fund datasets.**
- **Historical Data Warehousing for maintaining and analyzing long-term NAV records, fund metadata, and market trends.**
- **Advanced Financial Analytics including CAGR, rolling returns, volatility metrics, risk scores, Sharpe ratios, and performance benchmarking.**
- **Portfolio Intelligence & Risk Assessment through holdings overlap detection, diversification analysis, concentration risk evaluation, and sector exposure insights.**
- **Professional Reporting & Data Accessibility through automated generation of CSV, Excel, and PDF reports for research, analysis, and decision support.**

Ultimately, AMFIIntel Terminal aspires to evolve beyond a traditional mutual fund dashboard into a scalable fintech intelligence platform that empowers users to conduct comprehensive investment research, uncover portfolio risks, and make informed financial decisions with confidence.

System Architecture



Technology Stack

Frontend

Next.js

Used for:

- Server-side rendering
- Routing

TypeScript

Provides:

- Type safety

Tailwind CSS

Used for:

- Consistent styling
- Rapid UI development

ShadCN UI

Provides:

- Accessible interfaces
- Enterprise design system

Recharts

Used for:

- NAV visualizations
- Performance analytics

Backend

FastAPI

Acts as the central API gateway.

Responsible for:

- Authentication
- Fund APIs
- Analytics APIs
- Reporting APIs
- Portfolio APIs

SQLAlchemy

Used as ORM layer.

Provides:

- Database abstraction
- Query optimization
- Relationship management

APScheduler

Handles:

- Daily synchronization jobs
- Historical backfills
- Automated refreshes

Database Layer

PostgreSQL

Used as the primary data warehouse.

Stores:

Fund Metadata

- AMC Information

- Scheme Information
- Categories
- Benchmarks

Historical NAV Records

Daily NAV history for all schemes.

Asset Allocation

Fund allocation across:

- Equity
- Debt
- Cash
- Gold

Holdings Repository

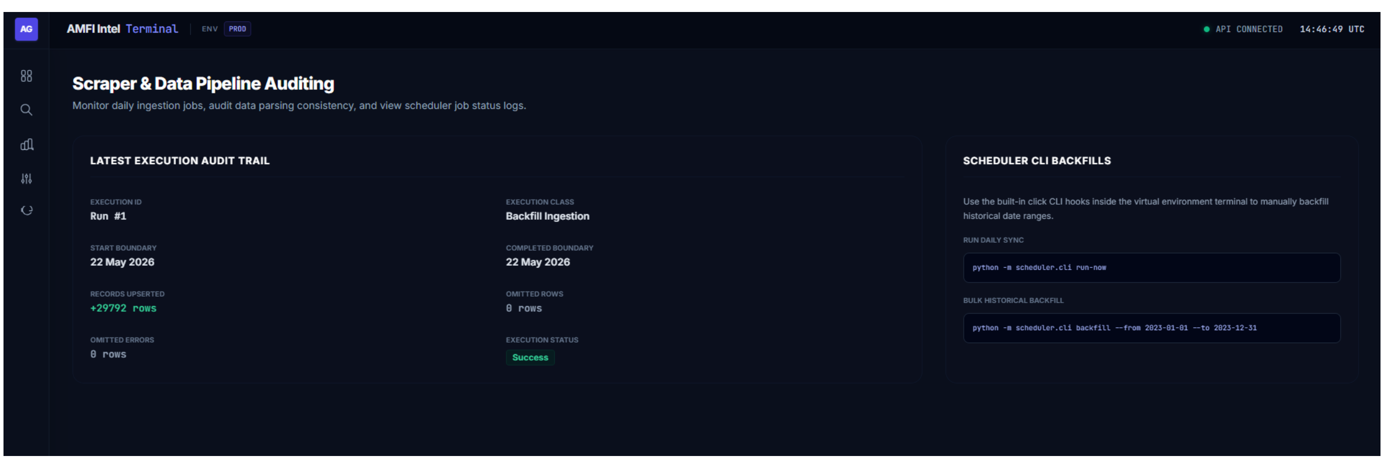
Stores:

- Company Holdings
- Sector Data
- Allocation Percentages

Analytics Cache

Stores:

- Rankings
- CAGR Metrics
- Risk Scores



Data Engineering Pipeline

Step 1 — Data Extraction

The scraper collects:

- Scheme metadata
- Daily NAV records
- Historical NAV datasets
- AUM disclosures

Step 2 — Validation

The system validates:

- Missing values
- Invalid records
- Duplicate entries

before processing.

Step 3 — Transformation

Raw AMFI data is normalized into a structured format.

This includes:

- Date standardization
- Category mapping
- Numeric formatting

Step 4 — Analytics Generation

The analytics engine computes:

CAGR

Measures long-term annualized growth.

Rolling Returns

Evaluates consistency over time.

Volatility

Measures return fluctuations.

Sharpe Ratio

Evaluates risk-adjusted performance.

Drawdown

Measures downside exposure.

Step 5 — Database Synchronization

Validated records are inserted into PostgreSQL using optimized bulk operations.

Dashboard Modules

Module 1 — Market Overview

Provides a real-time snapshot of the mutual fund ecosystem.

Displays:

- Tracked Assets
- Fund Houses
- Asset Classes
- NAV Freshness
- Top Yield Performers

Purpose:

Enable users to quickly understand current market conditions.

Module 2 — Mutual Fund Asset Explorer

Core research workspace.

Supports:

- Fund Discovery
- Search
- Filtering
- Category Analysis

- Risk Evaluation

Users can explore thousands of mutual funds through a structured interface.

Module 3 — NAV Query Engine

Provides direct access to historical NAV records.

Supports:

- Date Range Queries
- AMC Filters
- Category Filters

Enables detailed historical analysis.

Module 4 — AUM Intelligence Dashboard

Analyzes Assets Under Management across:

- Fund Categories
- Fund Houses
- Direct vs Regular Plans

Provides market-wide capital allocation insights.

Module 5 — Fund Comparison Engine

Allows users to compare multiple funds simultaneously.

Comparison Metrics:

- CAGR
- NAV Growth
- Volatility
- Sharpe Ratio
- Drawdown

Provides side-by-side investment evaluation.

Module 6 — Portfolio Lab

One of the flagship features.

Users can simulate mutual fund combinations.

The system computes:

Overlap Risk

Detects common holdings.

Diversification Score

Measures portfolio diversity.

Sector Exposure

Analyzes allocation concentration.

Concentration Risk

Identifies hidden portfolio vulnerabilities.

Module 7 — Pipeline Auditing Console

Provides transparency into data ingestion operations.

Tracks:

- Scheduler Jobs
- Backfills
- Records Processed
- Errors
- Execution Logs

This module improves operational reliability and monitoring.

Reporting Engine

The reporting system generates:

CSV Exports

Raw analytical datasets.

Excel Reports

Structured spreadsheet reports.

PDF Reports

Investor-grade research documents.

Supported Reports:

- Fund Reports
- Portfolio Reports
- Comparison Reports
- Historical Trend Reports

Security & Authentication

Implemented using:

JWT Authentication

Provides secure API access.

Password Hashing

Protects user credentials.

Role-Based Access

Supports:

- Admin
- Analyst
- Investor

roles.

Key Achievements

- Automated collection of AMFI mutual fund data.
- Built a complete financial data engineering pipeline.
- Designed a scalable PostgreSQL analytical warehouse.
- Developed a FastAPI-based service architecture.
- Implemented portfolio overlap intelligence.
- Created a Bloomberg-inspired fintech dashboard.
- Developed professional reporting capabilities.
- Enabled historical research and comparative analytics.

Conclusion

AMFIIntel Terminal transforms publicly available mutual fund information into a comprehensive investment intelligence ecosystem. By integrating automated data engineering, financial analytics, portfolio intelligence, historical research capabilities, and modern visualization technologies, the platform provides investors and analysts with a powerful environment for conducting data-driven mutual fund research and portfolio evaluation.

The project demonstrates expertise in full-stack development, data engineering, backend architecture, financial analytics, database design, system automation, and modern dashboard engineering, making it a production-grade fintech application rather than a conventional academic project.